737. Se os ângulos de vértices O_1, O_2, O_3, O_4 e O_5 medem, respectivamente, *90°*, *72°*, *135°*, *120°* e *105°* e os raios das circunferências de centros nestes vértices medem, respectivamente, *18 cm*, *35 cm*, *24 cm*, *36 cm* e *48 cm*, determine o comprimento da linha cheia *AB*.

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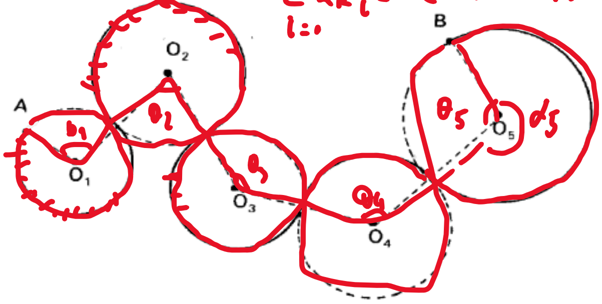
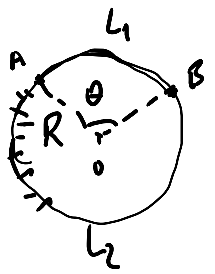
$$\sum_{i=1}^{5} 2R_i = 2(R_1+R_2+R_3+R_4+R_5)$$





$L_1 = \widehat{AB}$

$$\left.\begin{array}{l} 2\pi R \;—\; 2\pi \\ L_1 \;—\; \theta \end{array}\right\} \rightarrow \cancel{2\pi} L_1 = \cancel{2\pi} R\theta$$

$$\boxed{L_1 = R\theta}$$

$$\alpha = 2\pi - \theta \Rightarrow L_2 = R\alpha = R(2\pi - \theta)$$

$$\{R_1, R_2, R_3, R_4, R_5\} = \{18, 35, 24, 36, 48\}$$

$$\{\theta_1, \theta_2, \theta_3, \theta_4, \theta_5\} = \{90°, 72°, 135°, 120°, 105°\}$$

$$\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5\} = \{270°, 288°, 225°, 240°, 255°\}$$

$$\boxed{L_{AB} = R_1\alpha_1 + R_2\alpha_2 + R_3\alpha_3 + R_4\theta_4 + R_5\alpha_5}$$

$$L_{AB} = 18\times 270° + 35\times 288° + 24\times 225° + 36\times 240° + 48\times 255°$$

$$\left.\begin{array}{l} 180° \;—\; \pi \\ \theta° \;—\; \theta_{RAD} \end{array}\right\} \rightarrow \theta_{RAD} = \frac{\theta\,\pi}{180°}$$

$$L_{AB} = \underline{\pi}\,(18\times 270° + 35\times 288° + 24\times 225 + 36\times 240 + 48\times 255)$$

$$L_{AB} = \frac{\pi}{180}\left(19\times 270^\circ + 35\times 288^\circ + 24\times 225 + 36\times 240 + 48\times 255\right)$$

$$\boxed{L_{AB} = 205\,\pi}$$

N circ. DE RAIOS R_i , angulos θ_i , $i = 1, 2, \dots, N$

$$L_{AB} = \sum_{i=1}^{N} R_i\,\theta_i$$

$$\boxed{L_{AB} = \sum_{i=\{\theta_i\}}^{k} R_i\,\theta_i + \sum_{i:\{\alpha_i\}}^{m} R_i\,\alpha_i}$$